Code No. 3101

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (EEE, ECE, CSE AI&ML) I-Semester (MAIN) Examinations JAN/FEB-2025

PROGRAMMING FOR PROBLEM SOLVING

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

		Marks	BT	CO
1. a)	What is an algorithm, mention the key features of an algorithm.	2	1	1

b)	#include <stdio.h>
	Int main()
	{
	Int i=0, j=1;
	Printf("\n %d", i++ && ++j);
	Printf("\n %d %d", i, j);
	Return 0;
	}
	
	Give the output.
		2	3	2

c)	What is an array. Write the characteristics of an array.	2	1	3
d)	Mention the difference between strcat() and strncat().	2	2	4
e)	Differences between Array name and Pointer.	2	2	3
f)	List the four functions that can be used to read text files from the disk and write text files into the disk.	2	1	4
2. a)	Elaborate on compiling and executing high level language programs.	6	2	1
b)	Draw a flowchart to find the roots of a quadratic equation.	6	3	2
3. a)	Construct a C program that takes age as input from user and quits when a -1 is entered.	6	3	3
b)	Compare Recursive function with iterative function.	6	4	4

(P.T.O.)

4. a)	Define a preprocessor macro swap(t,x,y) that will swap two arguments x and y of a given type t.	6	3	3
b)	Sort an array LIST of size N and find the position of the target value T mention the algorithm.	6	3	2
5. a)	Write a program in C which accepts one line of text and prints the words in reverse order.	6	3	3
b)	Discuss the standard facilities to allocate and de-allocate dynamic heap memory in detail.	6	2	1
6. a)	What is FILE. How many files can be opened at once and what happens if the file is not closed.	6	2	4
b)	How do you assign a structure to another explain with an example.	6	3	4
7. a)	Discuss storage class specifiers for variables.	6	2	3
b)	How is a pointer variable declared. What is the purpose of the data type included in the declaration.	6	2	3

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a, C, b d = Sqrt(b*b-4

d<0 d>0

Δ

d = b^2 - 4ac d = -b/2a, -Δ/4a.

= -b ± √Δ / 2a = ±b/2a, Δ/4a.